

StudyAbroad.AI — Software Requirements Specification (SRS)

IEEE 830 Architecture, Data Models, Multi-Agent Orchestration & Security

System: StudyAbroad.AI Platform

Architecture: 12 Autonomous Agents

Distribution: Confidential Client Copy

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

StudyAbroad.AI — Autonomous Study Abroad Intelligence Platform

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1. Introduction

1.1 Purpose

This SRS defines the complete functional and non-functional requirements for StudyAbroad.AI, a fully autonomous AI-powered study abroad intelligence platform. Intended for the development team, AI engineers, project managers, and stakeholders.

1.2 Scope

StudyAbroad.AI is a web and mobile application that automates the entire study abroad journey for international students from initial research through university application, visa processing, and post-arrival integration. The system employs 12 autonomous AI agents, a multi-source RAG pipeline, and a full-stack web + mobile application.

The system WILL:

- Autonomously scrape and maintain data from 600+ universities across 20+ countries
- Monitor 100+ scholarship sources in real time
- Generate personalized, voice-preserved application documents
- Guide students through visa applications with country-specific knowledge
- Provide career ROI analytics and post-graduation employment data
- Support students from research phase through post-arrival life abroad

The system WILL NOT:

- Guarantee admission to any university
- Submit applications on behalf of students without explicit consent
- Store or share personal student data without GDPR-compliant consent
- Provide legally binding immigration advice

1.3 Definitions and Abbreviations

Term	Definition
SRS	Software Requirements Specification
RAG	Retrieval-Augmented Generation
LLM	Large Language Model
HITL	Human-in-the-Loop
SOP	Statement of Purpose
LOR	Letter of Recommendation
IELTS	International English Language Testing System
GRE	Graduate Record Examination
GMAT	Graduate Management Admission Test
PR	Permanent Residency
OPT	Optional Practical Training (USA)
PGWP	Post-Graduation Work Permit (Canada)
PSW	Post-Study Work (UK)
AES	Advanced Encryption Standard
GDPR	General Data Protection Regulation
JWT	JSON Web Token
SSR	Server-Side Rendering
SEO	Search Engine Optimization
CI/CD	Continuous Integration / Continuous Deployment

2. Overall Description

2.1 Product Perspective

StudyAbroad.AI is a standalone SaaS platform integrating multiple external data sources and AI services:

- Next.js frontend (web)
- React Native mobile application (iOS + Android)
- FastAPI backend (REST APIs)
- Multi-agent AI layer (LangGraph + CrewAI)
- Multiple databases (PostgreSQL, ChromaDB/Pinecone, Redis, TimescaleDB)

2.2 Product Functions (High-Level)

- FR-1: User registration, authentication, profile management
- FR-2: Autonomous university data scraping and indexing
- FR-3: Autonomous scholarship hunting and monitoring
- FR-4: AI-powered university and program recommendation
- FR-5: Voice-preserved application document generation
- FR-6: Professor outreach email automation
- FR-7: Multi-channel deadline tracking and alerting
- FR-8: Country-specific visa guidance and simulation
- FR-9: Career ROI and employment outcome analytics
- FR-10: Scholarship scam detection and validation
- FR-11: Financial document preparation assistance
- FR-12: Post-arrival wellbeing and integration support
- FR-13: 24/7 AI counselor with persistent memory
- FR-14: Human-in-the-Loop trust escalation
- FR-15: Multi-language platform support (10 languages)

2.3 User Classes

Primary: International Student (Prospective)

- Age 18–35, applying for Masters/PhD/Bachelors programs abroad
- Moderate technical literacy; uses smartphones and web
- Pain points: research overload, document prep, scholarships, visa confusion

Secondary: Parent/Sponsor

- Read-only view of student's application progress

Tertiary: Human Counselor (Expert)

- Reviews AI-escalated complex queries

Admin: Platform Administrator

- Full system access, scraping monitoring, user management

2.4 Operating Environment

- Web: Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
- Mobile: iOS 14+, Android 10+
- Deployment: Vercel (frontend), Railway (backend + agents)
- Screen Sizes: 320px (mobile) to 4K (desktop)

2.5 Design Constraints

- Must operate within free/low-cost API tiers for MVP
- Web scraping must comply with robots.txt and rate limits
- LLM token consumption must be optimized
- Platform must be GDPR-compliant from day one

3. System Features and Functional Requirements

3.1 Authentication and Profile Module

3.1.1 Registration and Auth

- REQ-AUTH-001: Users may register via email and password
- REQ-AUTH-002: One-click registration via Google OAuth 2.0
- REQ-AUTH-003: One-click registration via LinkedIn OAuth 2.0 with profile import
- REQ-AUTH-004: Email verification link sent upon registration
- REQ-AUTH-005: Password must be minimum 8 chars, 1 uppercase, 1 number, 1 special char
- REQ-AUTH-006: JWT session management with 7-day expiry
- REQ-AUTH-007: Refresh token support for seamless session renewal

3.1.2 Student Profile Builder

- REQ-PROFILE-001: Multi-step profile wizard with progress indicator

Academic Section:

- REQ-PROFILE-002: Collect highest degree, institution, graduation year, GPA/grade
- REQ-PROFILE-003: Accept GPA in all formats: 4.0 scale, 10-point, percentage, CGPA, UK classification
- REQ-PROFILE-004: Auto-convert any GPA format to standardized 4.0 scale
- REQ-PROFILE-005: Allow transcript upload (PDF/JPG/PNG); AI auto-extracts GPA and grades via OCR + LLM
- REQ-PROFILE-006: Collect test scores: IELTS, TOEFL, GRE, GMAT, LSAT, MCAT, SAT, DELF, TestDaF
- REQ-PROFILE-007: Track planned retake dates for test improvements

Personal and Financial:

- REQ-PROFILE-008: Collect nationality, study level, field of study, preferred start date
- REQ-PROFILE-009: Collect annual budget range with currency selection
- REQ-PROFILE-010: Collect preferred countries (multi-select), intake preference, program duration preference
- REQ-PROFILE-011: Collect career goals (target job title/industry post-graduation)
- REQ-PROFILE-012: Import work and education experience from LinkedIn

Research Background (PhD):

- REQ-PROFILE-013: Allow research interests entry (free-text + keyword tags)
- REQ-PROFILE-014: Allow upload of published papers or research reports
- REQ-PROFILE-015: Research experience description field

Profile Intelligence:

- REQ-PROFILE-016: Calculate Profile Strength Score (0-100) based on completeness and competitiveness
 - REQ-PROFILE-017: Identify profile gaps with specific improvement actions
 - REQ-PROFILE-018: What-If Simulator: adjust a score, see which new universities unlock
 - REQ-PROFILE-019: Peer benchmark: compare profile against historical admitted student data
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3.2 University Discovery Module

- REQ-UNIV-001: Natural language search interface (e.g. 'affordable CS Masters in Europe under 12000 euros')
 - REQ-UNIV-002: Parse natural language queries to extract: field, degree, country, budget, language of instruction
 - REQ-UNIV-003: Ranked results with relevance explanations
 - REQ-UNIV-004: Filter refinement: country, ranking, tuition, language, intake, acceptance rate, duration
 - REQ-UNIV-005: Program-level data (specific Masters/PhD programs, not just university level)
 - REQ-UNIV-006: Display per program: name, country, city, tuition, IELTS/GRE minimums, GPA min, deadline, acceptance rate, ranking, duration, intakes
 - REQ-UNIV-007: Side-by-side comparison of up to 5 universities across all data fields
 - REQ-UNIV-008: Cost of living per city (Numbeo integration)
 - REQ-UNIV-009: Student review aggregates (Reddit, TheStudentRoom, GMAT Club)
 - REQ-UNIV-010: Alumni employment outcomes per program
 - REQ-UNIV-011: Safety and quality of life scores per city
 - REQ-UNIV-012: Save universities to shortlist with notes and tags
 - REQ-UNIV-013: Subject-specific rankings (CSRankings, Shanghai Subject)
 - REQ-UNIV-014: Intake availability tracker (Fall/Spring/Summer/Rolling)
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3.3 AI Recommendation Engine

- REQ-REC-001: Generate personalized recommendations on profile completion
 - REQ-REC-002: Use RAG over vectorized university database for semantic profile matching
 - REQ-REC-003: Categorize into Dream (under 30% probability), Target (30–65%), Safety (65%+)
 - REQ-REC-004: Display match score (0–100%) with written explanation per university
 - REQ-REC-005: Estimate admission probability based on profile vs. university averages
 - REQ-REC-006: Full cost-of-attendance breakdown: tuition + living + visa + travel = total annual cost
 - REQ-REC-007: Include career ROI as a weighted ranking factor
 - REQ-REC-008: Suggest universities student may not have considered
 - REQ-REC-009: Recommend optimal intake based on application readiness
 - REQ-REC-010: Auto-update recommendations when student profile changes
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3.4 Scholarship and Funding Hub

- REQ-SCHOL-001: Personalized scholarship list matched to profile, field, nationality, target countries
- REQ-SCHOL-002: Pre-check eligibility before displaying (filter ineligible scholarships)
- REQ-SCHOL-003: Display per scholarship: name, coverage, eligible nationalities, GPA requirements, deadline, competition level, official URL
- REQ-SCHOL-004: Prioritize hidden gem scholarships (less competitive, high relevance)
- REQ-SCHOL-005: Show scholarship success rate per profile type

- REQ-SCHOL-006: Funding gap calculator: total cost minus all matched scholarships
 - REQ-SCHOL-007: Part-time work regulations per country (max hours, average wages)
 - REQ-SCHOL-008: Student loan comparison by country (lenders, rates, moratoriums)
 - REQ-SCHOL-009: Flag potentially fraudulent scholarships
 - REQ-SCHOL-010: Scholarship negotiation guide and email templates
 - REQ-SCHOL-011: Anonymized scholarship essay bank for reference
 - REQ-SCHOL-012: Push/email alerts when new matching scholarships open
 - REQ-SCHOL-013: TA/RA funded position finder within target programs
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3.5 Document Preparation Suite

Voice Preservation System:

- REQ-DOC-001: 10-question story-prompt session to capture student writing style
- REQ-DOC-002: Extract vocabulary level, sentence structure, tone, narrative style from responses
- REQ-DOC-003: All generated documents reflect captured voice profile

Statement of Purpose:

- REQ-DOC-004: Unique SOP per university + program + professor combination
- REQ-DOC-005: Incorporates: why this university, why this program, academic background, career goals
- REQ-DOC-006: Enforces word limits per university requirements (500–1000 words typically)
- REQ-DOC-007: Quality score on 5 dimensions: Clarity, Specificity, Motivation, University Fit, Uniqueness (0–10 each)
- REQ-DOC-008: Actionable improvement suggestions per dimension
- REQ-DOC-009: One-click re-tailoring from one university to another in under 60 seconds

Other Documents:

- REQ-DOC-010: Personal Statement (UK/European style)
- REQ-DOC-011: Research Proposal (PhD): title, background, questions, methodology, timeline
- REQ-DOC-012: LOR draft templates for professors/employers
- REQ-DOC-013: LOR request email generator
- REQ-DOC-014: Academic CV builder (academic format, not corporate ATS)
- REQ-DOC-015: Diversity and Inclusion Statement (US universities)
- REQ-DOC-016: Scholarship-specific essays per scholarship prompt
- REQ-DOC-017: Waitlist Appeal Letter generator
- REQ-DOC-018: Deferral Request Letter generator

Quality Assurance:

- REQ-DOC-019: Plagiarism check with similarity score on all documents
- REQ-DOC-020: AI-detection score check (target below 40% AI probability)
- REQ-DOC-021: Grammar and clarity API check

Document Vault:

- REQ-DOC-022: Central vault for all student documents (upload, store, organize)
- REQ-DOC-023: Supported formats: PDF, DOCX, JPG, PNG (max 50MB per file)
- REQ-DOC-024: Version history for last 10 versions per document
- REQ-DOC-025: Share document via secure link with mentor, parent, counselor
- REQ-DOC-026: All vault documents encrypted with AES-256 at rest

3.6 Application Tracking Module

- REQ-APP-001: Kanban tracker with stages: Research, Drafting, Submitted, Under Review, Interview, Waitlisted, Offered, Accepted, Rejected
 - REQ-APP-002: Each university card shows: stage, next deadline, match score, document completion status
 - REQ-APP-003: University-specific document checklist (what to submit, format, portal links)
 - REQ-APP-004: Application fee tracker and total spend calculator
 - REQ-APP-005: Offer Letter Analyzer: AI extracts and explains conditions, scholarships, enrollment deadlines
 - REQ-APP-006: Multi-Offer Comparison: financial breakdown of 2+ offers side by side
 - REQ-APP-007: Admission Interview Prep: mock interview with AI feedback per question
 - REQ-APP-008: Auto-generated application timeline working backwards from each deadline
 - REQ-APP-009: Application success probability per university based on profile + documents
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3.7 Deadline Watcher Module

- REQ-DL-001: Track all deadlines: application, scholarship, financial aid, housing, visa, post-admission immigration
 - REQ-DL-002: Alerts via: in-app notification, email, WhatsApp (optional), Telegram (optional)
 - REQ-DL-003: Google Calendar and Outlook Calendar sync
 - REQ-DL-004: Weekly Monday task list: top 5 priority tasks this week
 - REQ-DL-005: Smart priority ordering by urgency and impact
 - REQ-DL-006: Reminder schedule: 30 days, 14 days, 7 days, 3 days, 1 day, day-of alerts
 - REQ-DL-007: Custom reminder frequency settings
 - REQ-DL-008: Countdown timers displayed on dashboard for all active deadlines
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3.8 Visa and Immigration Module

- REQ-VISA-001: Country-specific visa roadmaps for: USA F-1, UK Student Visa, Germany Studienvisum, Canada Study Permit, Australia Visa 500, Netherlands MVV, and 10+ others
 - REQ-VISA-002: Personalized document checklist per visa type and home country combination
 - REQ-VISA-003: Bank balance calculator using official country formula and current thresholds
 - REQ-VISA-004: Financial sponsorship letter templates
 - REQ-VISA-005: Embassy appointment booking guidance and processing time estimates
 - REQ-VISA-006: Visa Interview Mock Coach: AI plays visa officer, evaluates answers, gives feedback
 - REQ-VISA-007: Visa Rejection Recovery: identify likely rejection reason, create appeal or reapplication plan
 - REQ-VISA-008: Post-study immigration guides: OPT/H-1B, PSW, PGWP, EU Blue Card, PR pathways
 - REQ-VISA-009: Immigration policy change alerts for target countries
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3.9 Career ROI Module

- REQ-ROI-001: Employment outcomes per program: median salary 1/3/5yr, top employers, placement rate, internship availability
 - REQ-ROI-002: ROI score: salary boost / tuition cost = years to break even
 - REQ-ROI-003: Country job market health score per field
 - REQ-ROI-004: Best country for your field comparison tool
 - REQ-ROI-005: Career ROI as weighted factor in university recommendations
 - REQ-ROI-006: Job board filtered by visa work eligibility
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3.10 AI Counselor Module

- REQ-CHAT-001: Persistent chat interface accessible from any page
 - REQ-CHAT-002: Full access to student's profile, shortlist, documents, and application history as context
 - REQ-CHAT-003: Answers questions about universities, scholarships, visas, documents, careers, finances
 - REQ-CHAT-004: Persistent conversation memory across sessions
 - REQ-CHAT-005: Responds in student's preferred language (10 languages)
 - REQ-CHAT-006: Displays confidence level and cites sources for factual claims
 - REQ-CHAT-007: Flags low-confidence answers and offers human expert escalation
 - REQ-CHAT-008: Human expert response delivered within 24 hours for escalated queries
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3.11 Financial Documentation Module

- REQ-FIN-001: Bank balance calculation per student's visa type, home country, target country
 - REQ-FIN-002: Bank Statement Preparation Guide customized per home country banking system
 - REQ-FIN-003: Financial sponsorship declaration letter templates
 - REQ-FIN-004: Funding gap calculator: total cost minus scholarships and savings
 - REQ-FIN-005: Education loan comparison: lenders, interest rates, moratoriums, repayment
 - REQ-FIN-006: Currency exchange planning guidance
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3.12 Post-Arrival Life Abroad Module

- REQ-LIFE-001: Pre-Departure Checklist customized per destination country
- REQ-LIFE-002: First 30/60/90 Day Plan for each country
- REQ-LIFE-003: Cultural Adaptation Guide per destination country
- REQ-LIFE-004: Banking Setup Guide per country
- REQ-LIFE-005: Verified student housing integration (scam detection applied)
- REQ-LIFE-006: Community connector: same home country + same university students
- REQ-LIFE-007: Connect students in same program and intake year
- REQ-LIFE-008: Mental health and counseling resources per university
- REQ-LIFE-009: Anonymous weekly wellbeing check-in with stress detection

- REQ-LIFE-010: Part-time job board filtered by visa work hour allowances
 - REQ-LIFE-011: Post-graduation immigration filing deadline tracker (OPT, PGWP, PSW)
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3.13 Professor Outreach Module

- REQ-PROF-001: Professor database: name, department, research interests, recent publications, grants, lab openings, email, profile URL
 - REQ-PROF-002: Semantic matching: student research background vs professor research
 - REQ-PROF-003: Rank professors by research alignment score and funding probability
 - REQ-PROF-004: Personalized cold outreach email referencing specific professor paper or grant
 - REQ-PROF-005: 3-email follow-up sequence if no reply after 7 and 14 days
 - REQ-PROF-006: Track email status: drafted / sent / replied / interview scheduled
 - REQ-PROF-007: Research statement tailored to each professor's lab
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3.14 Scam Detection Module

- REQ-SCAM-001: Validate all scholarship URLs against whitelist of official scholarship domains
 - REQ-SCAM-002: Flag scholarships requiring application fees, banking details, or with no verifiable sponsor
 - REQ-SCAM-003: University accreditation verification against Ministry of Education lists
 - REQ-SCAM-004: Verify This feature: student pastes any URL or email; system outputs legitimacy score
 - REQ-SCAM-005: Community-reported scam database with warnings
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4. Non-Functional Requirements

4.1 Performance

- REQ-PERF-001: Web app initial page load under 2 seconds on 10 Mbps
- REQ-PERF-002: University search results under 3 seconds
- REQ-PERF-003: Recommendation list generated within 10 seconds of profile submission
- REQ-PERF-004: SOP first draft generated within 30 seconds
- REQ-PERF-005: Support 10,000 concurrent users without degradation
- REQ-PERF-006: API uptime 99.5% monthly
- REQ-PERF-007: Weekly scrape cycle completed within 6 hours
- REQ-PERF-008: Push notifications delivered within 60 seconds

4.2 Scalability

- REQ-SCALE-001: Database supports up to 1 million student profiles
- REQ-SCALE-002: Vector database supports up to 5 million embedded chunks
- REQ-SCALE-003: Horizontal scaling for FastAPI backend
- REQ-SCALE-004: Parallelized scraping pipeline (1000+ concurrent workers)

4.3 Reliability

- REQ-REL-001: Automatic retry with exponential backoff (maximum 3 retries) for failed agent tasks
- REQ-REL-002: Graceful degradation: show cached results if AI service unavailable
- REQ-REL-003: Database backups every 24 hours with 30-day retention
- REQ-REL-004: Recovery Time Objective (RTO): restore from backup within 4 hours
- REQ-REL-005: Recovery Point Objective (RPO): maximum 24 hours data loss

4.4 Usability

- REQ-UX-001: New user completes profile setup within 10 minutes
- REQ-UX-002: Primary actions reachable within 3 clicks from dashboard
- REQ-UX-003: Fully responsive from 320px to 4K resolution
- REQ-UX-004: WCAG 2.1 Level AA accessibility compliance
- REQ-UX-005: Error messages in plain language (no technical jargon)
- REQ-UX-006: Supports 10 languages: English, Arabic, Chinese Simplified, Hindi, Bengali, French, German, Spanish, Turkish, Urdu
- REQ-UX-007: Interactive tooltips and walkthrough for first-time users

4.5 Maintainability

- REQ-MAINT-001: All AI agent actions logged in Langfuse
- REQ-MAINT-002: Minimum 70% unit test coverage
- REQ-MAINT-003: Scraper configurations externalized to config files (not hardcoded)
- REQ-MAINT-004: Structured JSON logs for all API calls and errors
- REQ-MAINT-005: Automated deployment via GitHub Actions CI/CD
- REQ-MAINT-006: Sentry dashboard for real-time error tracking and alerting

4.6 Compliance

- REQ-COMP-001: GDPR (EU Regulation 2016/679) compliant
- REQ-COMP-002: Data deletion: student data deleted within 72 hours of request
- REQ-COMP-003: Data export: student data exported as JSON within 24 hours
- REQ-COMP-004: Cookie consent banner on first visit with consent logging
- REQ-COMP-005: Privacy policy accessible from all pages
- REQ-COMP-006: No student data used for AI training without explicit opt-in consent

5. External Interface Requirements

5.1 UI Requirements

- Web app: Next.js 14 App Router
- UI components: shadcn/ui + Tailwind CSS
- Animations: Framer Motion
- Color scheme: dark mode default, light mode available

- Typography: Inter font (Google Fonts)
- Real-time: Socket.io WebSocket

5.2 External APIs and Services

Service	Purpose
Google Gemini 1.5 Pro	Primary LLM for agent reasoning
Google Gemini Flash	Fast lightweight tasks
Google text-embedding-004	Vector embedding generation
Groq	Fallback LLM
Crawl4AI	AI-native web scraping
Playwright	Dynamic JS page scraping
Pinecone	Production vector database
ChromaDB	Development vector database
Twilio WhatsApp API	Deadline notifications via WhatsApp
Resend	Transactional email delivery
Cloudinary	Document storage and CDN
Numbeo	Cost of living data
Google Calendar API	Deadline calendar sync
LinkedIn OAuth 2.0	Profile data import
Google OAuth 2.0	User authentication
Langfuse	AI agent observability
LangSmith	LLM evaluation and testing
Sentry	Error tracking
PostHog	Product analytics

5.3 Communication Requirements

- All transit data encrypted via TLS 1.3 (HTTPS enforced)
- RESTful API design (FastAPI backend)
- WebSocket protocol via Socket.io for real-time features
- Message queues via Redis + Celery for internal agent communication
- JSON format for all API responses with consistent error structure

6. AI Agent Specifications

6.1 Framework

- Complex stateful workflows: LangGraph (state machine, cyclic loops, HITL checkpoints)
- Sequential pipelines: CrewAI (role-based assembly line agents)
- Hybrid: LangGraph orchestrates top-level; delegates pipeline tasks to CrewAI crews

6.2 Agent Execution Standards

- REQ-AGENT-001: All executions logged: agent name, input, output, tokens, execution time, cost, errors
- REQ-AGENT-002: Retry logic: maximum 3 retries with exponential backoff
- REQ-AGENT-003: Critical decisions pass through HITL trust layer before delivery
- REQ-AGENT-004: Confidence scores computed and displayed with all advice
- REQ-AGENT-005: Sources cited for all factual claims (URL, timestamp)

6.3 LLM Configuration Per Agent

Agent	Model	Temperature	Framework
UniversityScraperAgent	Gemini Flash	0.1	CrewAI
ScholarshipHunterAgent	Gemini Flash	0.1	CrewAI
RecommenderAgent	Gemini 1.5 Pro	0.3	LangGraph
DocumentCraftingAgent	Gemini 1.5 Pro	0.7	LangGraph
ProfessorOutreachAgent	Gemini 1.5 Pro	0.6	CrewAI
DeadlineWatcherAgent	Gemini Flash	0.1	LangGraph
VisaGuideAgent	Gemini 1.5 Pro	0.2	LangGraph
CounselorAgent	Gemini 1.5 Pro	0.5	LangGraph
CareerROIAgent	Gemini 1.5 Pro	0.2	CrewAI
ScamDetectorAgent	Gemini Flash	0.0	Rule-based + LLM
FinancialDocAgent	Gemini 1.5 Pro	0.2	LangGraph
WellbeingCompanionAgent	Gemini 1.5 Pro	0.6	LangGraph

6.4 Scraping Schedule

Agent	Frequency	Trigger
UniversityScraperAgent	Weekly (Sunday 2:00 AM UTC)	Cron + manual
ScholarshipHunterAgent	Daily (12:00 AM UTC)	Cron + manual
ProfessorOutreachAgent	On-demand	User action
CareerROIAgent	Monthly (1st of month)	Cron
ScamDetectorAgent	Real-time per new scholarship	Event trigger

7. Data Requirements

7.1 Core Database Entities (PostgreSQL)

Users: user_id (UUID PK), email, password_hash, name, created_at, subscription_tier, preferred_language, oauth_provider

StudentProfiles: profile_id, user_id (FK), degree_level, field_of_study, gpa_raw, gpa_scale, gpa_converted_40, ielts_score, toefl_score, gre_score, nationality, home_country, budget_min, budget_max, preferred_countries (JSON), career_goals, research_interests, profile_strength_score

Universities: university_id, name, country, city, qs_ranking, the_ranking, website, accreditation_status, last_scraped_at

Programs: program_id, university_id (FK), name, degree_level, field, language_of_instruction, tuition_international, duration_months, gpa_minimum, ielts_minimum, gre_minimum, acceptance_rate, application_deadline_fall, application_deadline_spring, intake_available (JSON), last_scraped_at

Scholarships: scholarship_id, name, funding_body, coverage_type, eligible_nationalities (JSON), eligible_fields (JSON), gpa_requirement, deadline, award_amount, official_url, is_verified, is_scam_flagged, last_scraped_at

Applications: application_id, user_id (FK), university_id (FK), program_id (FK), status (enum), applied_at, decision_at, offer_letter_url

Documents: document_id, user_id (FK), document_type (enum), title, content (encrypted), university_id (FK nullable), version, plagiarism_score, ai_detection_score, quality_score, created_at

Deadlines: deadline_id, user_id (FK), entity_type, entity_id, deadline_date, reminder_channels (JSON), is_completed

Professors: professor_id, university_id (FK), name, department, research_interests (JSON), recent_publications (JSON), has_open_position, email, last_scraped_at

OutreachEmails: email_id, user_id (FK), professor_id (FK), subject, body (encrypted), status (enum), sent_at, replied_at, follow_up_count

Conversations: conversation_id, user_id (FK), agent_name, messages (JSON encrypted), created_at, updated_at

7.2 Vector Database (ChromaDB / Pinecone)

university_programs collection: program_id as id, embedding of program description + requirements, metadata: university_name, country, field, tuition, gpa_min, intake

scholarships collection: scholarship_id as id, embedding of description + eligibility, metadata: name, deadline, coverage, eligible_countries

professor_profiles collection: professor_id as id, embedding of research interests + publication abstracts, metadata: university, department, email

visa_guides collection: guide_id as id, embedding of visa requirements, metadata: country, visa_type, last_updated

career_outcomes collection: program_id as id, embedding of career outcomes text, metadata: median_salary, placement_rate, top_employers

8. Security Requirements

- REQ-SEC-001: All data in transit encrypted via TLS 1.3 (HTTPS enforced)
 - REQ-SEC-002: All documents encrypted at rest using AES-256
 - REQ-SEC-003: Database backups encrypted using AES-256
 - REQ-SEC-004: Passwords hashed using bcrypt (minimum cost factor 12)
 - REQ-SEC-005: JWT tokens expire after 7 days; refresh tokens after 30 days
 - REQ-SEC-006: Rate limiting: maximum 100 API requests per minute per user
 - REQ-SEC-007: LLM API rate limiting: maximum 20 document generations per day on free tier
 - REQ-SEC-008: OWASP Top 10 mitigations: SQL injection prevention, XSS prevention, CSRF protection
 - REQ-SEC-009: Admin endpoints require multi-factor authentication
 - REQ-SEC-010: Log all authentication events (login, logout, failed attempts) with IP and timestamp
 - REQ-SEC-011: After 5 failed login attempts: account locked for 30 minutes
 - REQ-SEC-012: Documents accessible only by owner, explicitly invited collaborators, and admins (with audit log)
 - REQ-SEC-013: Scraping respects robots.txt; minimum 2-second delay between requests per domain
 - REQ-SEC-014: All API keys stored in environment variables only
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9. Constraints and Assumptions

9.1 Constraints

- CON-001: MVP built on free-tier APIs to minimize cost
- CON-002: Web scraping must comply with each website's terms of service
- CON-003: System cannot guarantee admission outcomes; all predictions are probabilistic
- CON-004: Visa guidance is informational only, not legal immigration advice
- CON-005: LLM context window limits require chunking strategy for large university data queries

9.2 Assumptions

- ASS-001: Target university websites remain publicly accessible for scraping
- ASS-002: Google Gemini API free tier available with sufficient quota for MVP
- ASS-003: Students have internet connectivity and a modern browser or mobile device
- ASS-004: Platform operates in English for MVP; multilingual added in Phase 7
- ASS-005: Initial data seed (200 universities, 500 scholarships) is sufficient for MVP launch
- ASS-006: Human expert counselors sourced within 3 months post-launch for HITL escalation

10. Appendix

Appendix A: Supported Countries (Scraping Scope)

USA, UK, Canada, Germany, Australia, Netherlands, Sweden, France, Switzerland, New Zealand, Ireland, Japan, South Korea, Singapore, UAE, Finland, Denmark, Italy, Spain, Belgium

Appendix B: Supported Languages

English, Arabic, Chinese Simplified, Hindi, Bengali, French, German, Spanish, Turkish, Urdu

Appendix C: Test Score Ranges

Test	Score Range
IELTS	0.0 - 9.0
TOEFL iBT	0 - 120
GRE	260 - 340
GMAT	200 - 800
LSAT	120 - 180
MCAT	472 - 528
SAT	400 - 1600

Appendix D: GPA Conversion

Scale	Excellent	Good	Satisfactory	Pass
4.0 Scale	3.7-4.0	3.0-3.6	2.5-2.9	2.0-2.4
10-Point	9.0-10.0	7.5-8.9	6.5-7.4	5.0-6.4
Percentage	90-100%	75-89%	65-74%	50-64%
UK Class	First	2:1	2:2	Third

